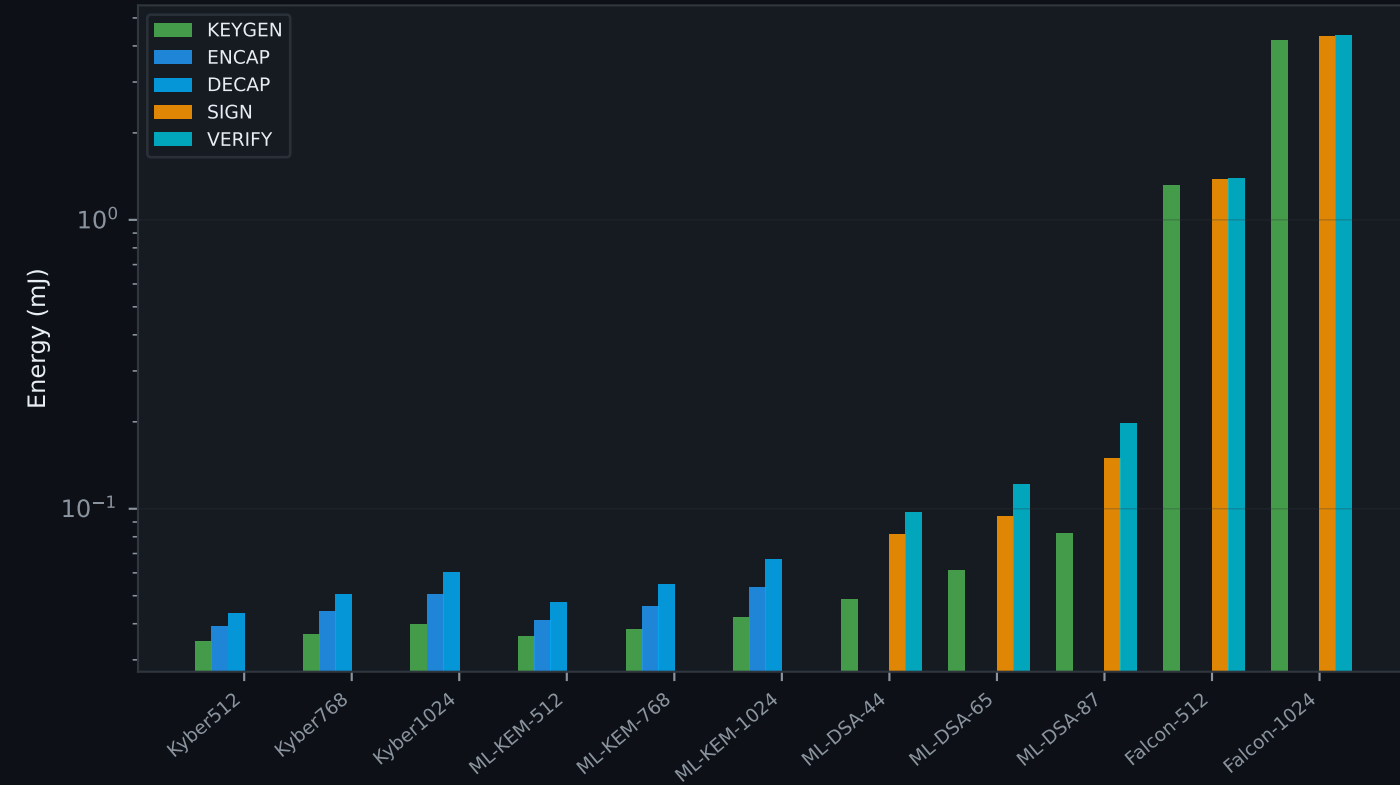
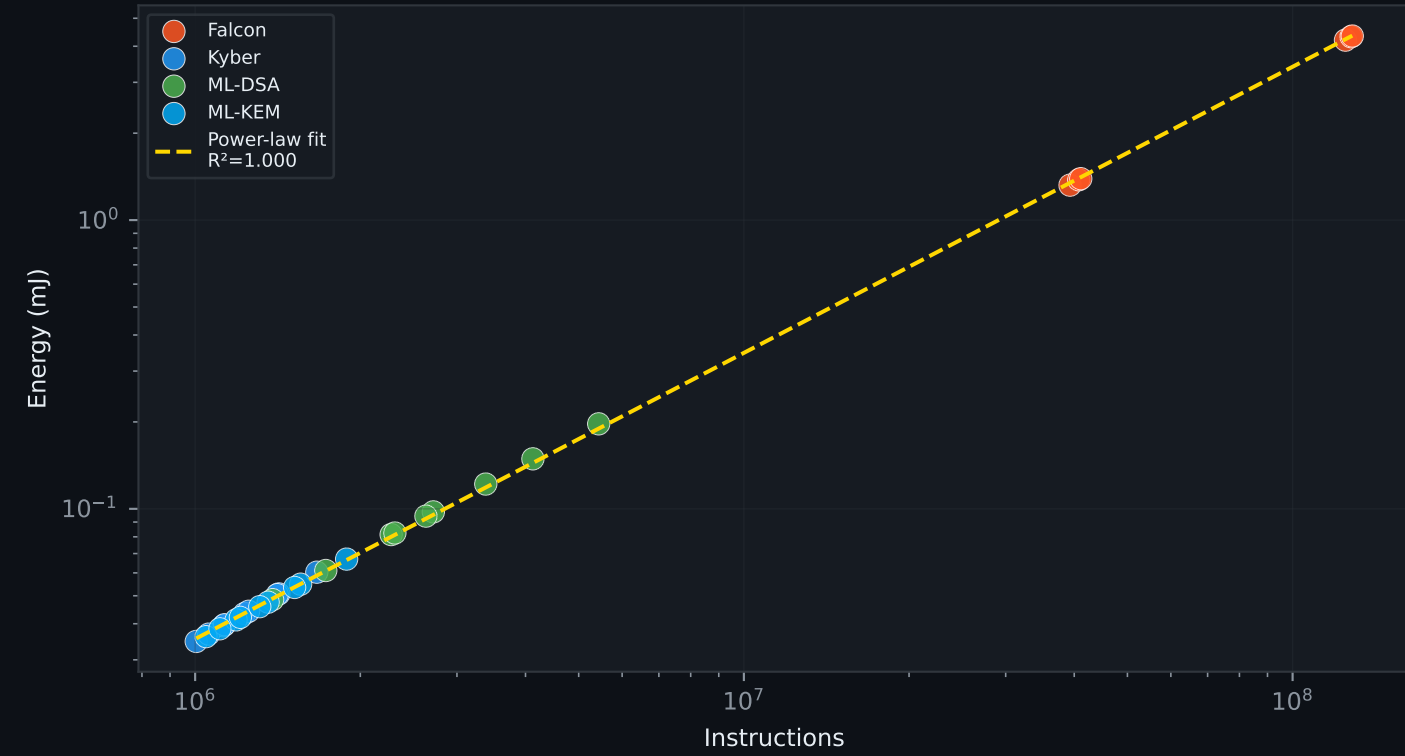


PQC Energy Usage — Real gem5 ARM Measurements
ARM Cortex-M4 | 64 MHz | 66 mW active | liboqs

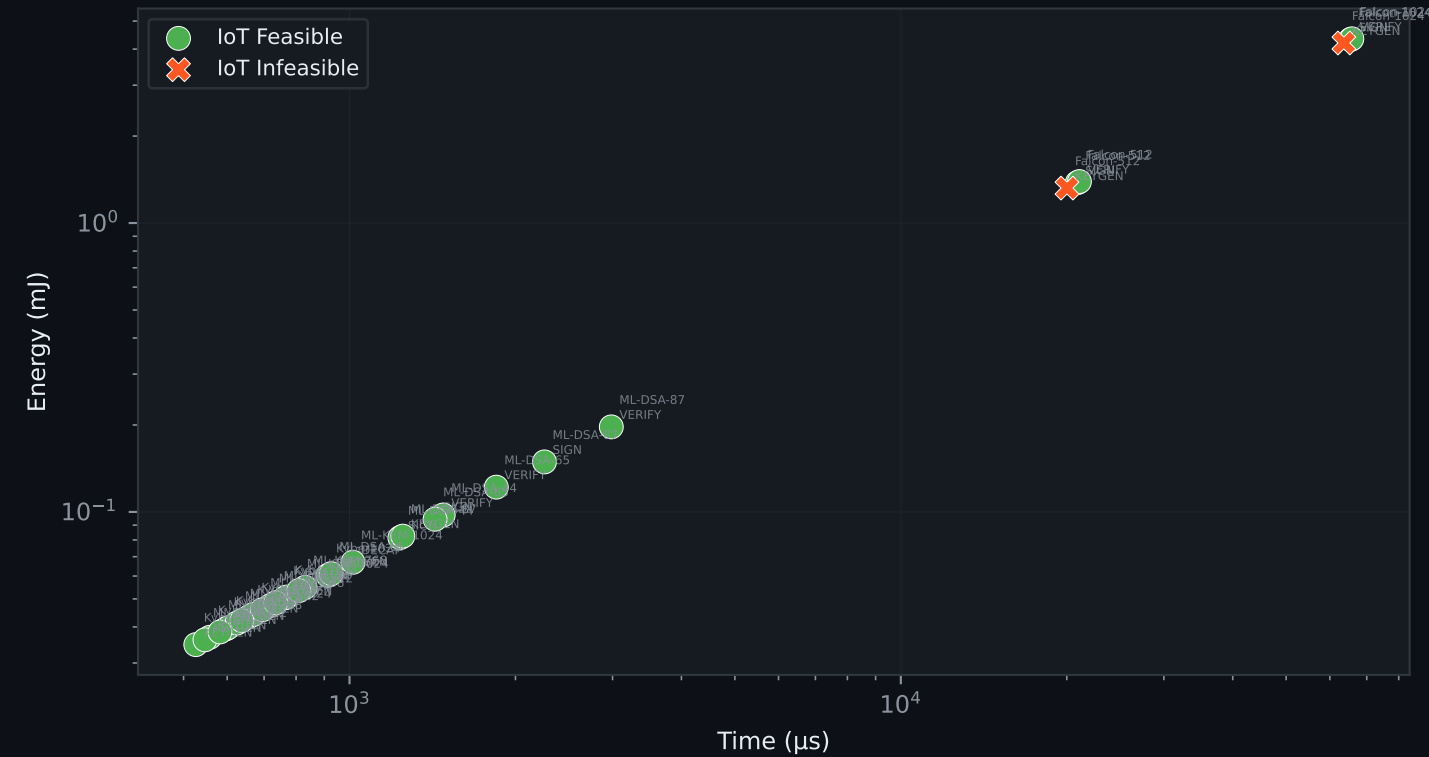
A — Energy per Algorithm & Operation



B — Instructions vs Energy (log-log)



C — IoT Feasibility Map
(KEYGEN <10ms, ops <100ms)

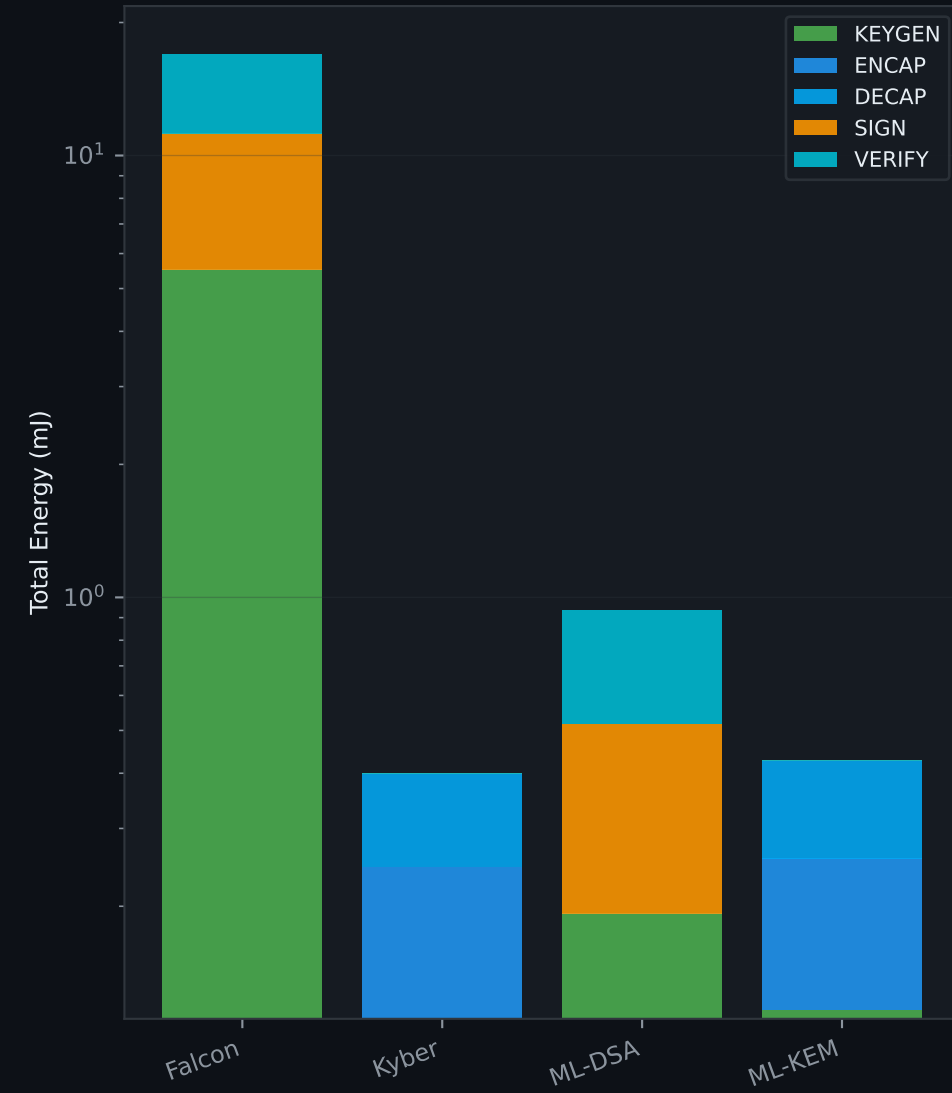


D — Energy Distribution by Algorithm Family

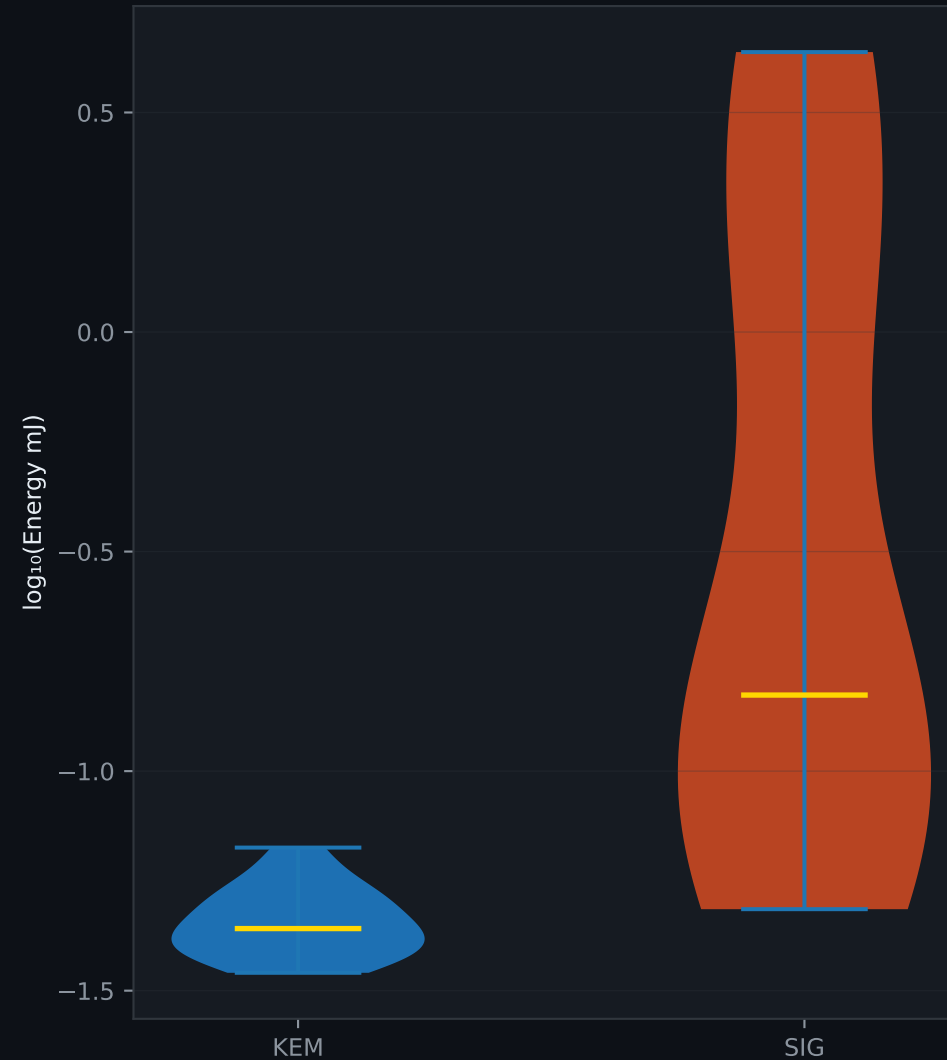


Energy Breakdown — KEM vs SIG Families

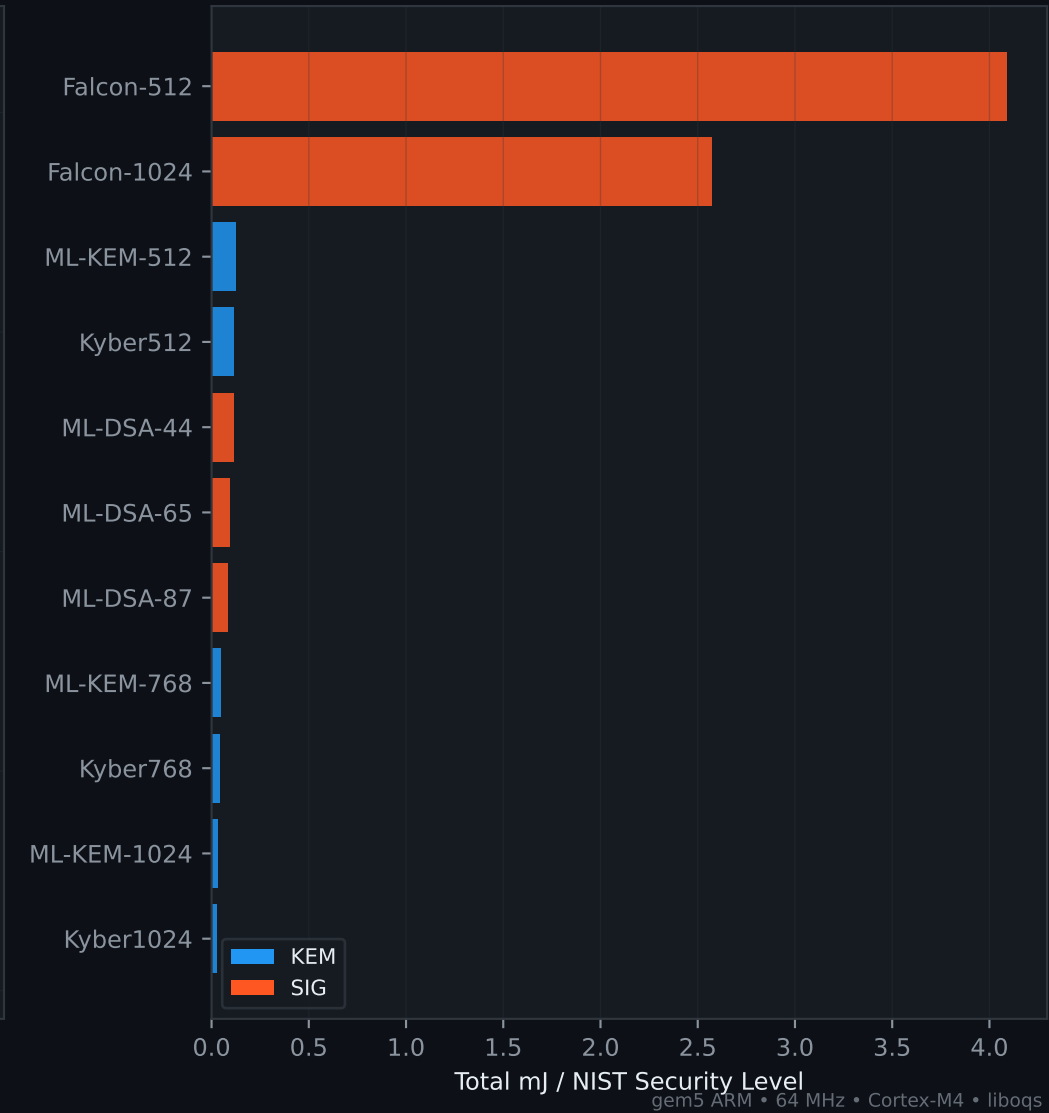
A — Total Energy by Family & Op



B — KEM vs SIG Distribution

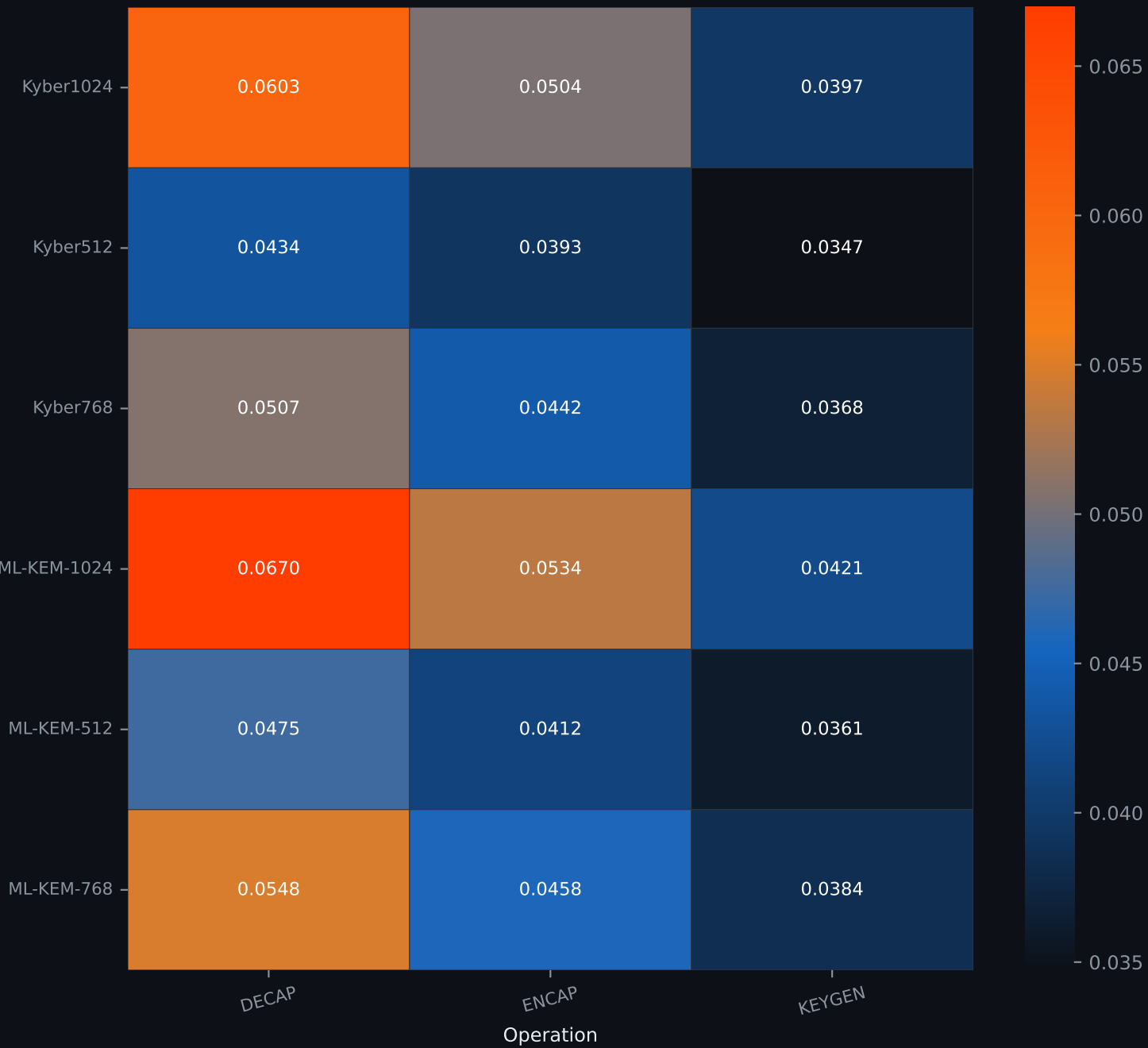


C — Energy Efficiency (lower = better per security bit)



Energy Heatmaps (mJ) — gem5 ARM Measurements

KEM Algorithms

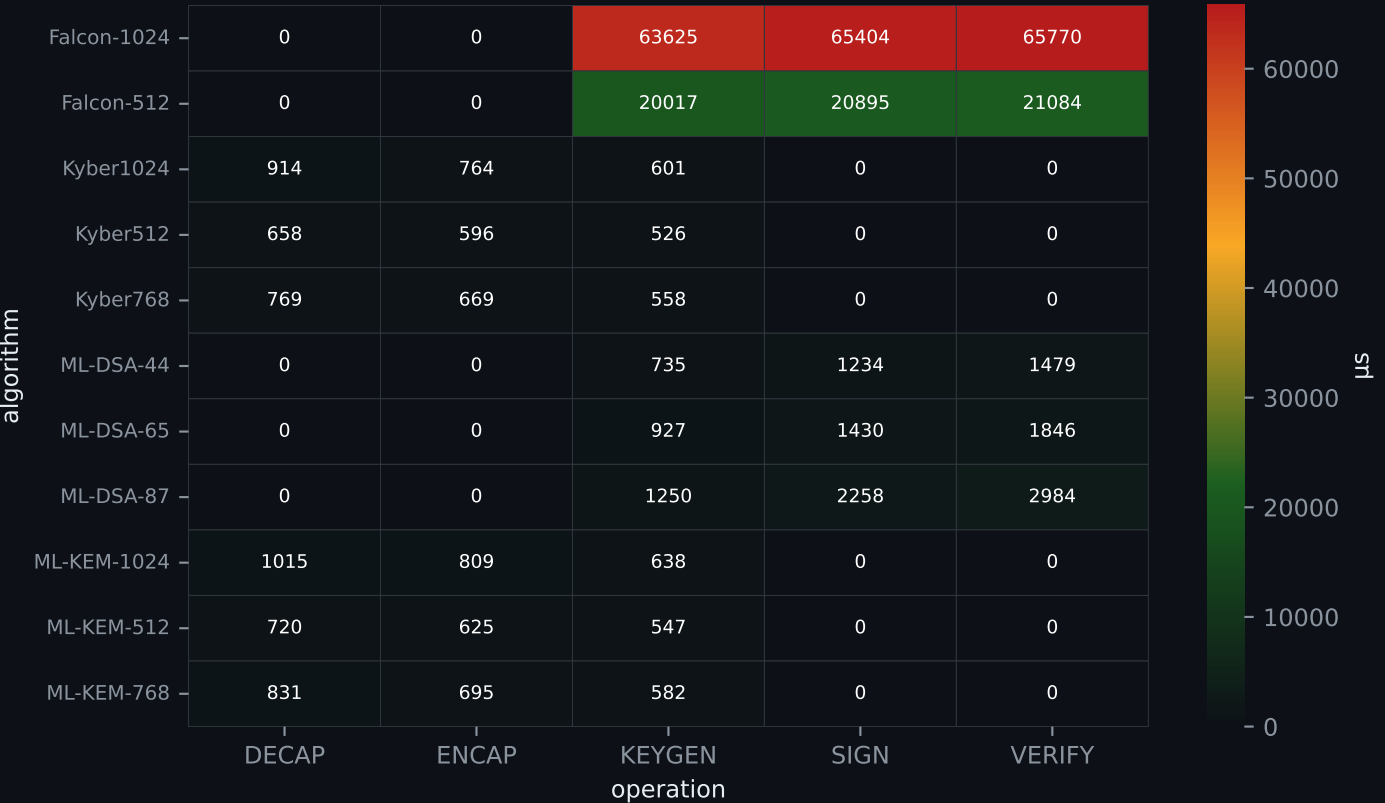


SIG Algorithms

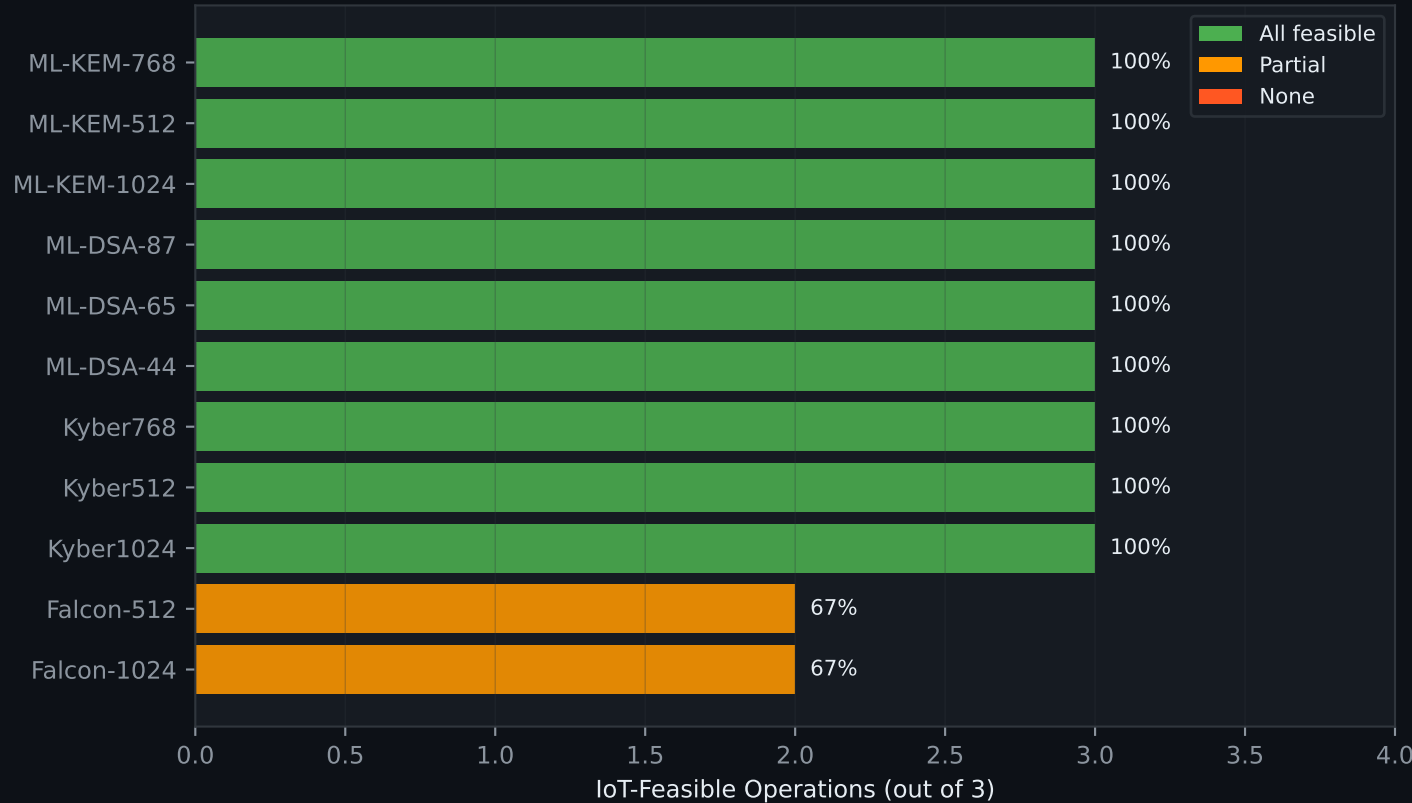


IoT Feasibility Analysis — ARM Cortex-M4 @ 64 MHz

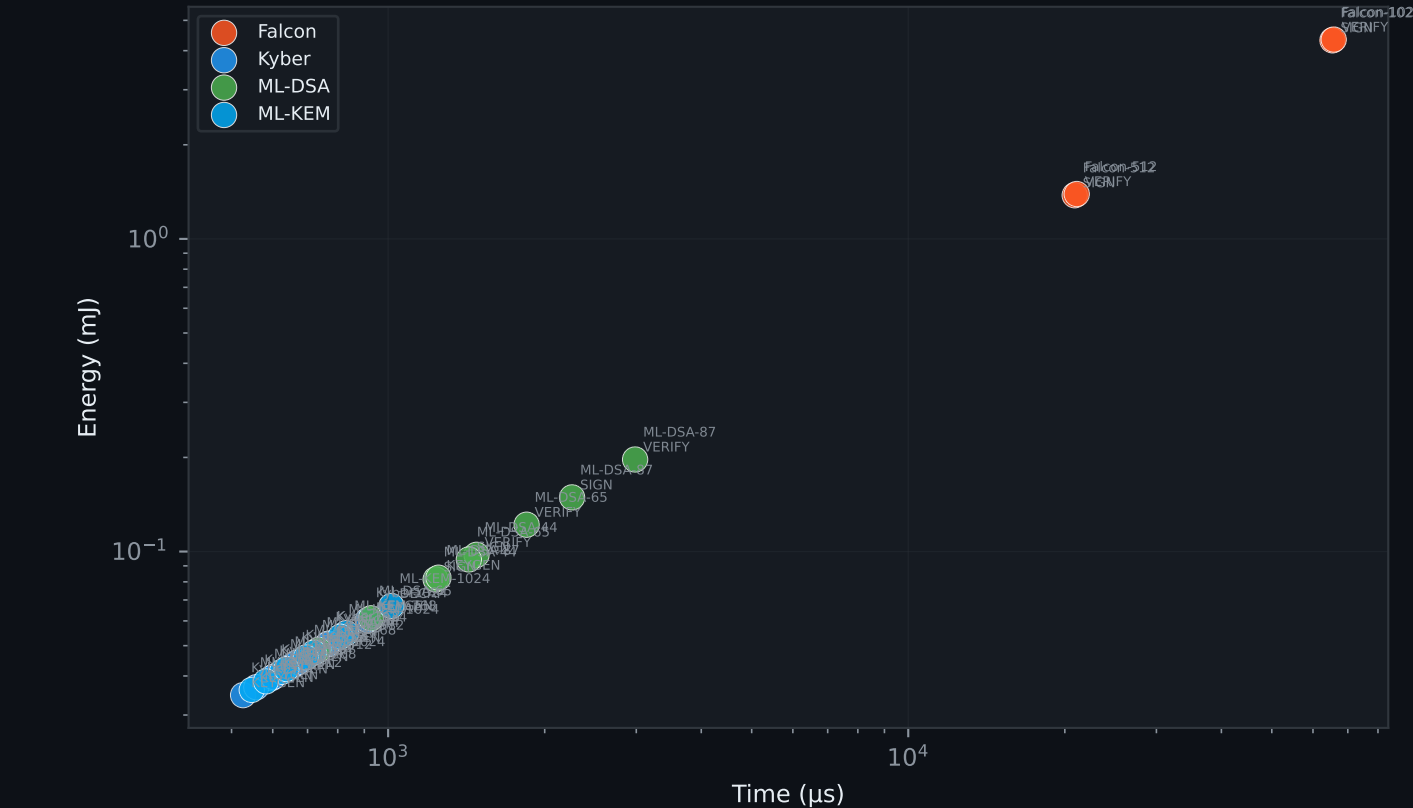
A — Execution Time (μs)



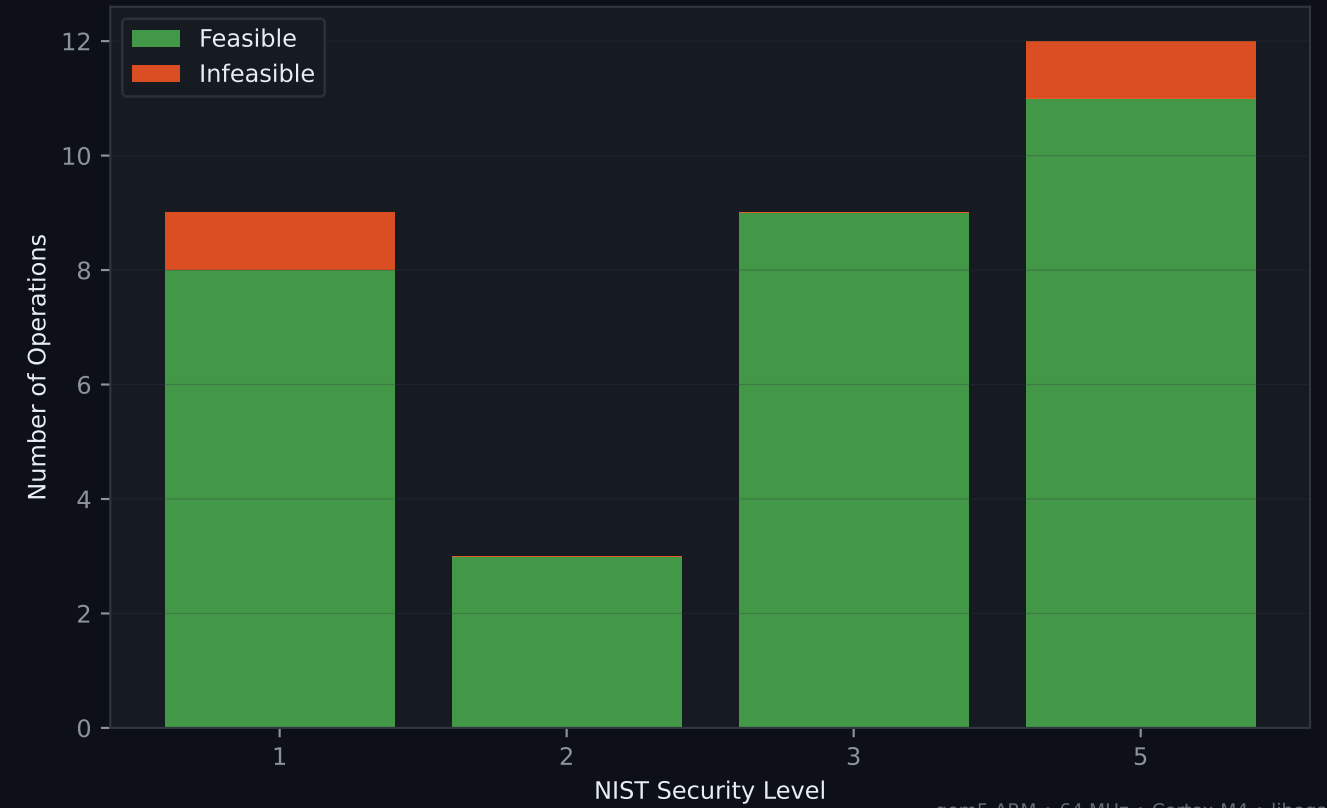
B — IoT-Feasible Ops per Algorithm



C — IoT-Feasible Operations Only

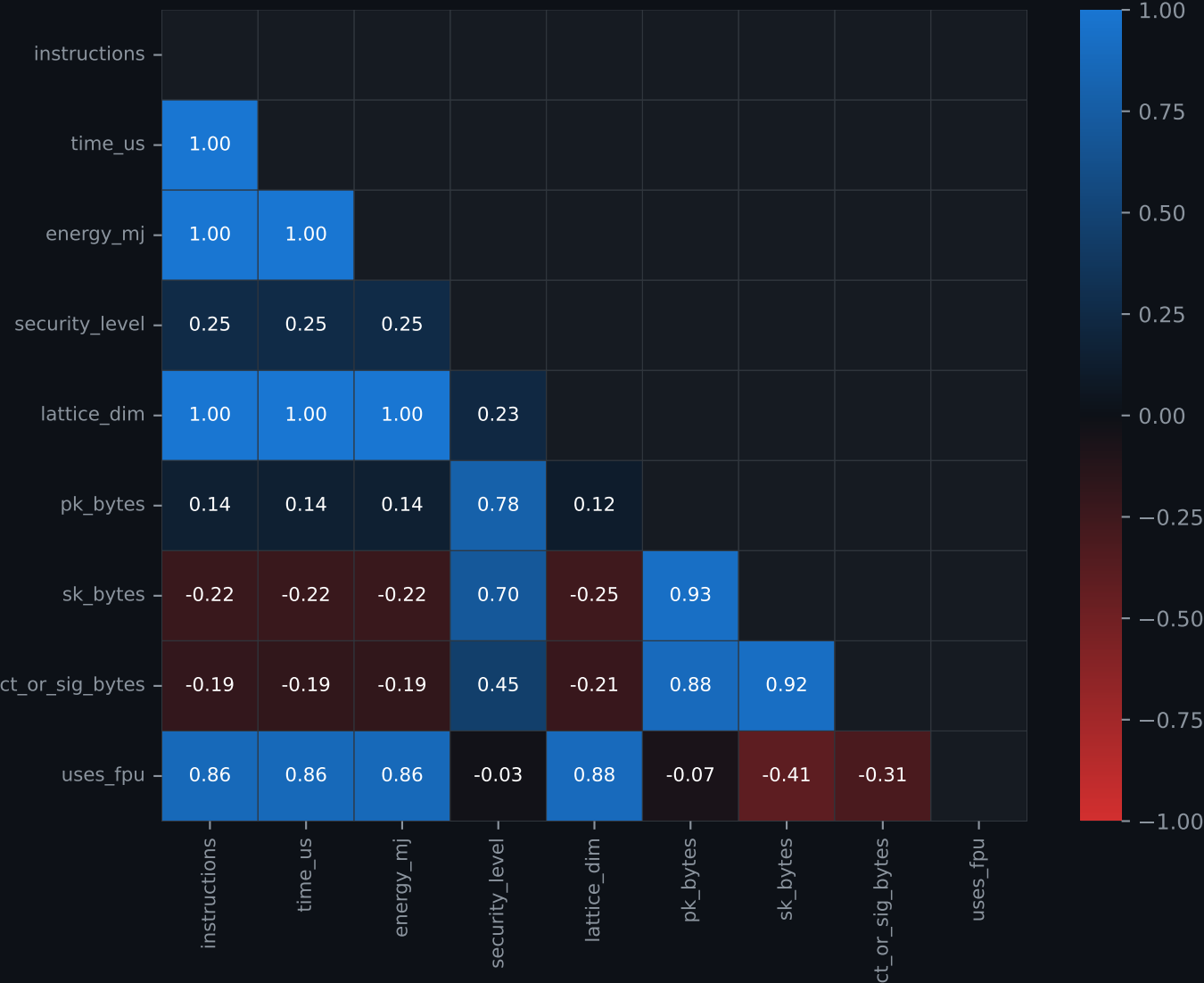


D — IoT Feasibility by Security Level

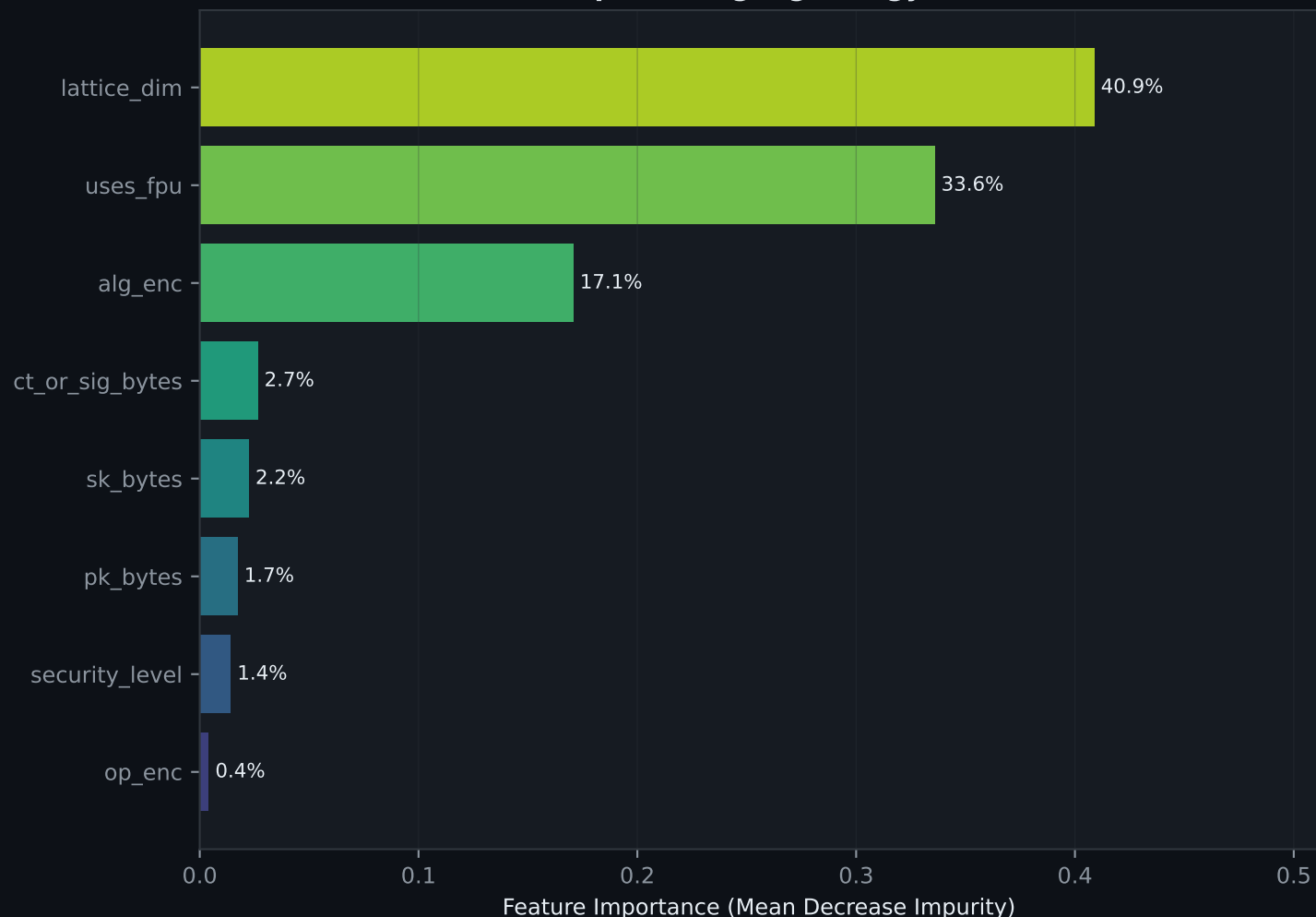


Feature Correlation & Importance for Energy Prediction

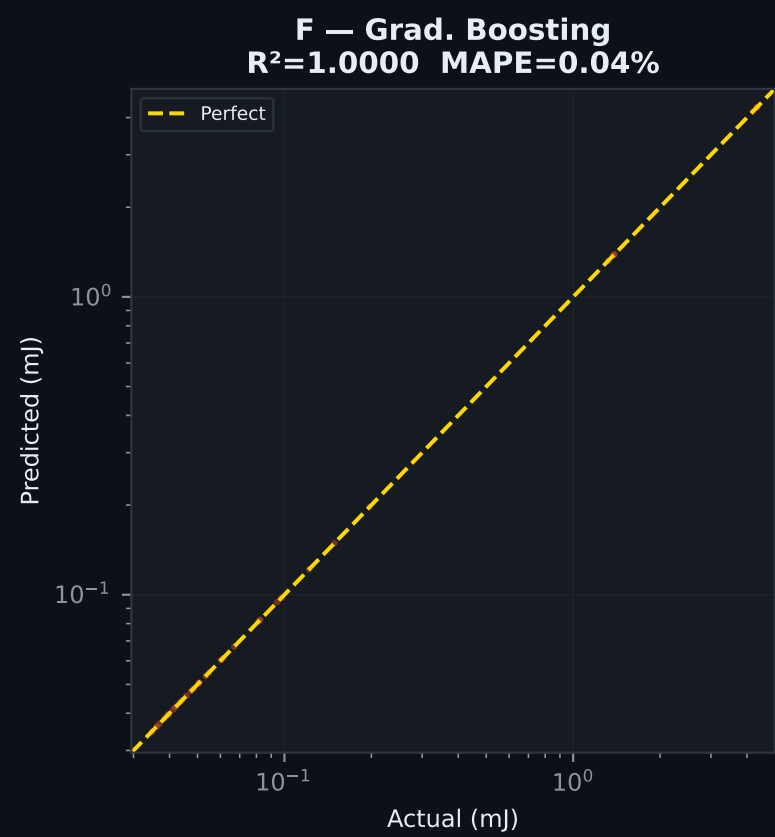
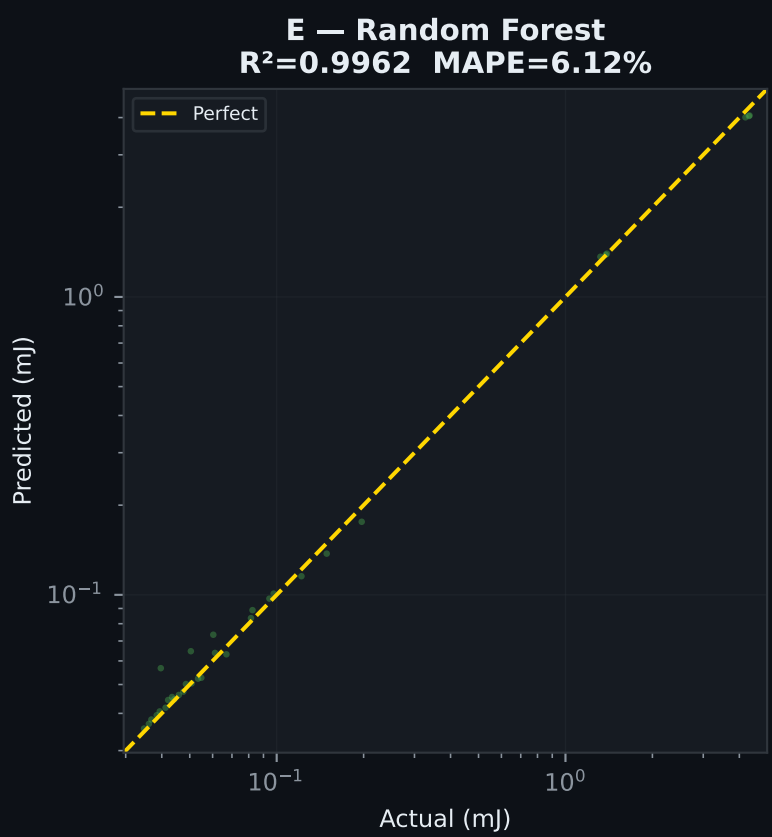
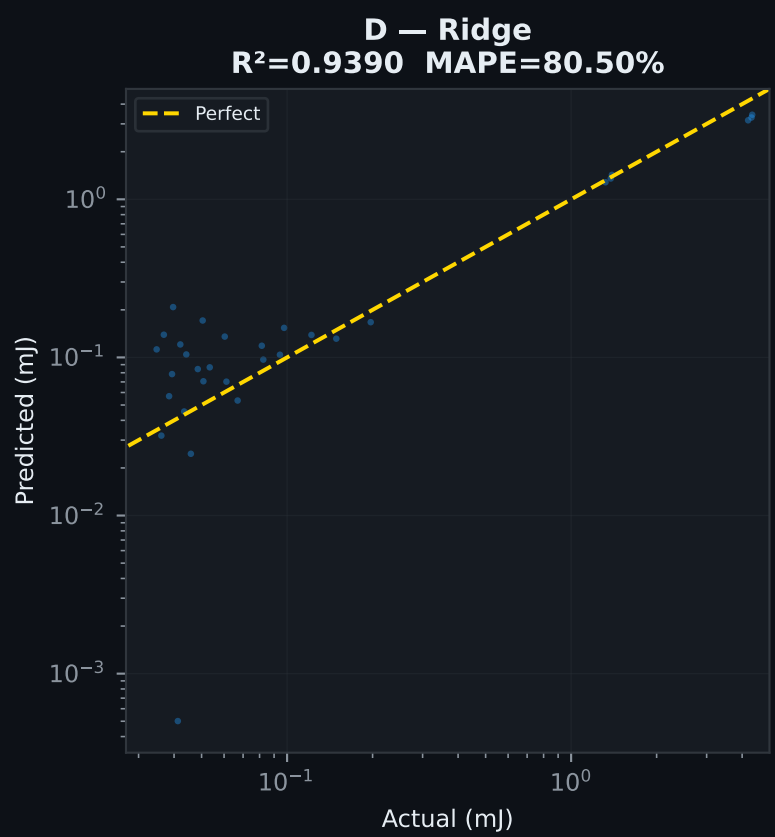
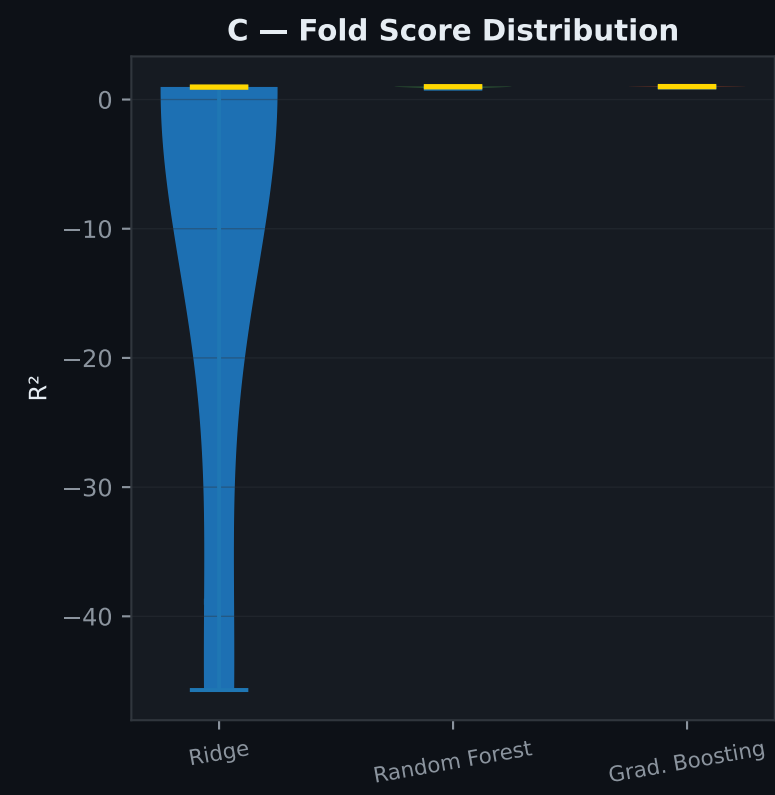
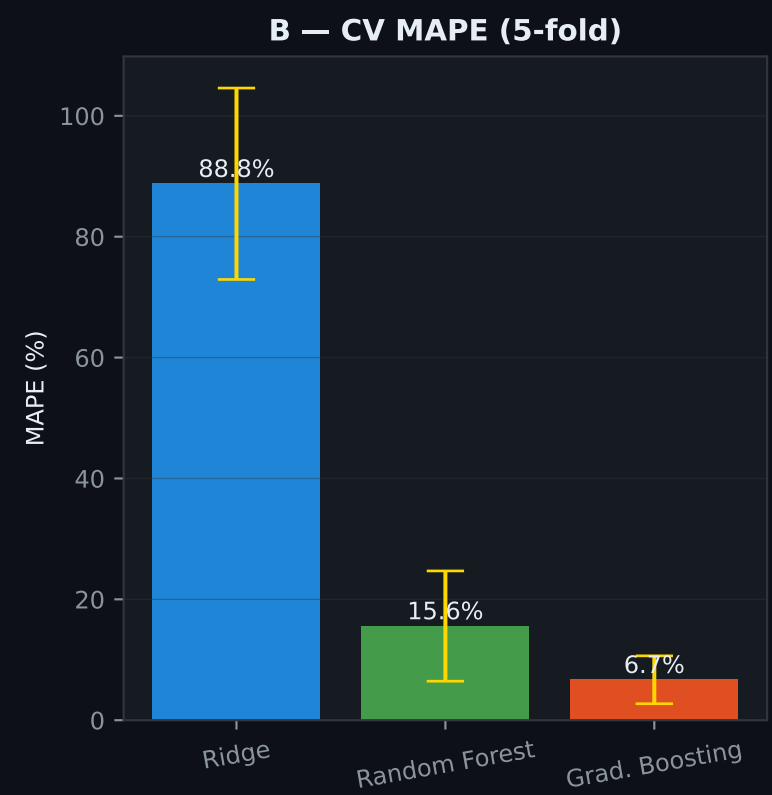
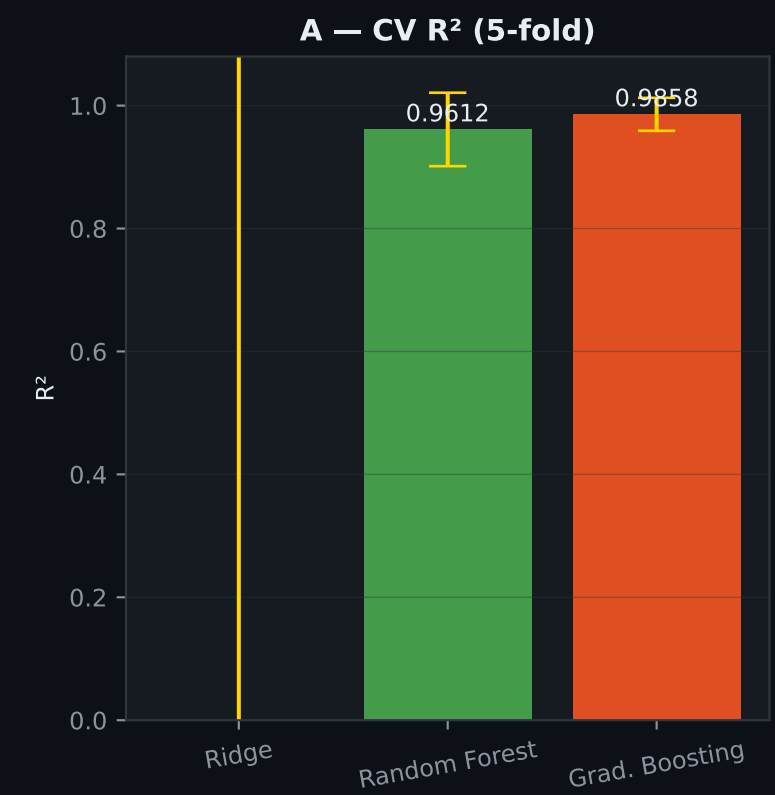
A — Pearson Correlation Matrix



B — Random Forest Feature Importance
(predicting log energy)



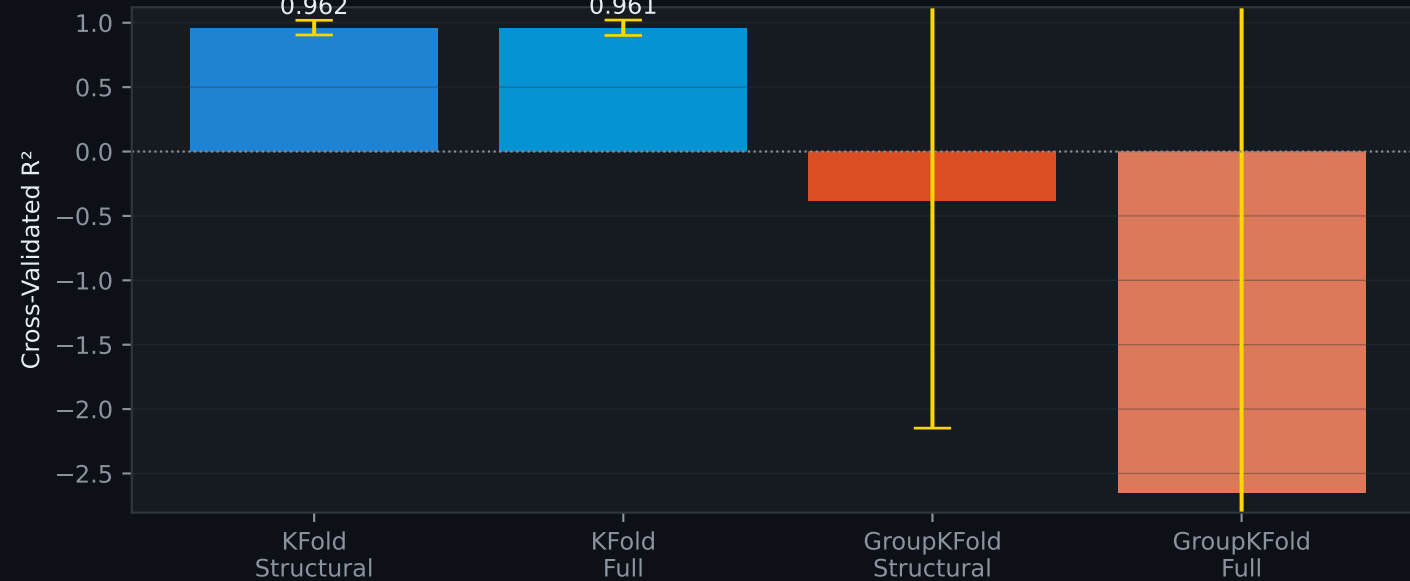
ML Energy Prediction — 5-Fold Cross Validation
Training on 100-run expanded dataset (3,300 rows)



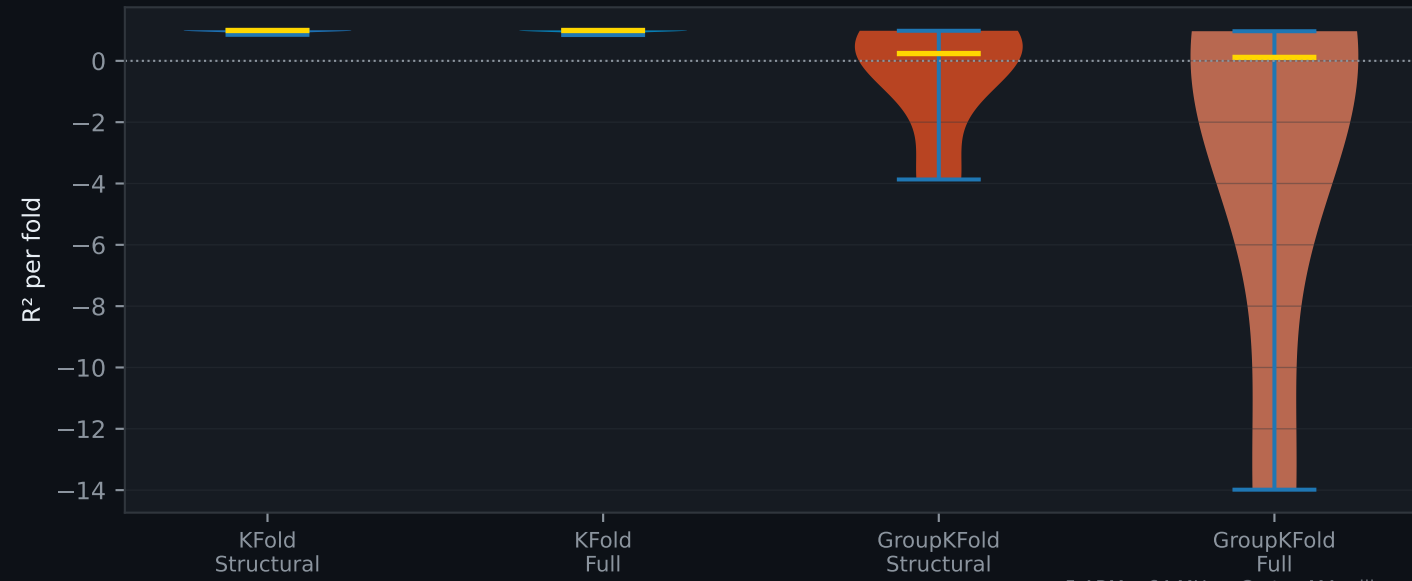
Experiment B — Cross-Algorithm Generalisation (GroupKFold)
Tests whether model predicts unseen algorithms from structure alone

-2.654

**A — KFold vs GroupKFold
(Structural vs Full features)**



**B — R^2 Fold Distribution
(width = variance)**



Summary — gem5 ARM Results & ML Model Performance

A — All gem5 Measurements (33 rows)

Algorithm	Op	Instructions	Time (µs)	Energy (mJ)	IoT
Kyber1024	DECAP	1,666,035	914.1	0.0603	✓
Kyber1024	ENCAP	1,411,418	764.4	0.0504	✓
Kyber1024	KEYGEN	1,131,129	601.4	0.0397	✓
Kyber512	DECAP	1,230,556	657.9	0.0434	✓
Kyber512	ENCAP	1,125,732	596.2	0.0393	✓
Kyber512	KEYGEN	1,004,680	526.2	0.0347	✓
Kyber768	DECAP	1,420,996	768.9	0.0507	✓
Kyber768	ENCAP	1,251,368	669.3	0.0442	✓
Kyber768	KEYGEN	1,058,582	557.9	0.0368	✓
ML-KEM-1024	DECAP	1,888,189	1014.7	0.0670	✓
ML-KEM-1024	ENCAP	1,517,961	809.4	0.0534	✓
ML-KEM-1024	KEYGEN	1,209,103	637.6	0.0421	✓
ML-KEM-512	DECAP	1,358,552	719.7	0.0475	✓
ML-KEM-512	ENCAP	1,187,481	624.9	0.0412	✓
ML-KEM-512	KEYGEN	1,046,730	546.8	0.0361	✓
ML-KEM-768	DECAP	1,556,727	830.8	0.0548	✓
ML-KEM-768	ENCAP	1,311,387	694.5	0.0458	✓
ML-KEM-768	KEYGEN	1,109,614	582.2	0.0384	✓
Falcon-1024	KEYGEN	124,847,496	63625.1	4.1993	✗
Falcon-1024	SIGN	127,775,399	65404.3	4.3167	✓
Falcon-1024	VERIFY	128,457,057	65770.1	4.3408	✓
Falcon-512	KEYGEN	39,296,848	20017.4	1.3211	✗
Falcon-512	SIGN	40,755,364	20895.4	1.3791	✓
Falcon-512	VERIFY	41,104,846	21083.7	1.3915	✓
ML-DSA-44	KEYGEN	1,383,960	734.6	0.0485	✓
ML-DSA-44	SIGN	2,276,270	1234.3	0.0815	✓
ML-DSA-44	VERIFY	2,717,016	1479.0	0.0976	✓
ML-DSA-65	KEYGEN	1,729,955	926.5	0.0612	✓
ML-DSA-65	SIGN	2,633,678	1429.6	0.0944	✓
ML-DSA-65	VERIFY	3,385,397	1846.1	0.1218	✓
ML-DSA-87	KEYGEN	2,311,900	1249.5	0.0825	✓
ML-DSA-87	SIGN	4,125,784	2257.7	0.1490	✓
ML-DSA-87	VERIFY	5,437,585	2984.4	0.1970	✓

B — ML Results & IoT Feasibility

Model	CV R² Mean	± Std	CV MAPE	Note
Ridge	-8.3709	± 18.6673	88.76%	
Random Forest	0.9612	± 0.0598	15.58%	
Grad. Boosting	0.9858	± 0.0268	6.69%	★ Best

IoT Feasibility Summary

✓ Kyber5123/3 ops feasible

✓ Kyber7683/3 ops feasible

✓ Kyber10243/3 ops feasible

✓ ML-KEM-5123/3 ops feasible

✓ ML-KEM-7683/3 ops feasible

✓ ML-KEM-10243/3 ops feasible

✓ ML-DSA-443/3 ops feasible

✓ ML-DSA-653/3 ops feasible

✓ ML-DSA-873/3 ops feasible

~ Falcon-5122/3 ops feasible

~ Falcon-10242/3 ops feasible